

08. DDPM & Stable Diffusion

Overview

Denoising Diffusion Probabilistic Models (DDPM) implement discrete-time reverse steps with learned noise prediction. We cover the simplified training loss, sampler schedules, classifier-free guidance, and how Stable Diffusion moves diffusion to a VAE latent space for efficiency.

Lab: Lab 08.

Prior modules: Module 07.

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Module track

This module builds on **Module 07**. Read those PDFs first. References to other modules appear as **Module NN** (not page numbers, because each module is its own PDF).



Figure 1: Module sequence (solid box: this PDF; dashed: typical next step).

1 DDPM Formulation

Denoising Diffusion Probabilistic Models [1] simplify the reverse variance and reparameterize mean as noise prediction. Builds on forward process **Module 07** (forward diffusion).

2 Simplified Training Loss

Ho et al. show a weighted denoising score matching objective:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} \left[\|\epsilon - \epsilon_{\theta}(\mathbf{x}_t, t)\|^2 \right], \quad (1)$$

where $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$, $\epsilon \sim \mathcal{N}(0, \mathbf{I})$, $t \sim \text{Uniform}(1, T)$.

Note. Predict noise ϵ rather than $\mathbf{x}_0$ directly, numerically stable and equivalent under reparameterization.

3 Sampling Algorithm

Starting from $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$, iterate for $t = T, \dots, 1$:

$$\mu_{\theta}(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(\mathbf{x}_t, t) \right), \quad (2)$$

$$\mathbf{x}_{t-1} = \mu_{\theta} + \sigma_t \mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(0, \mathbf{I}) \text{ if } t > 1. \quad (3)$$

4 U-Net Denoiser Backbone

ϵ_{θ} is a U-Net with timestep embedding (sinusoidal, as in **Module 04** (transformer block)). Skip connections preserve spatial detail, architecture shared with segmentation in **Module 14** (U-Net architecture).

5 Stable Diffusion: Latent Space

Stable Diffusion applies diffusion in VAE latent $\mathbf{z}$ rather than pixel space, reducing compute:

$$\mathcal{L} = \mathbb{E} \left[\|\epsilon - \epsilon_{\theta}(\mathbf{z}_t, t, \mathbf{c})\|^2 \right], \quad (4)$$

where $\mathbf{c}$ is text conditioning via cross-attention (transformer blocks from **Module 04**).

6 Classifier-Free Guidance

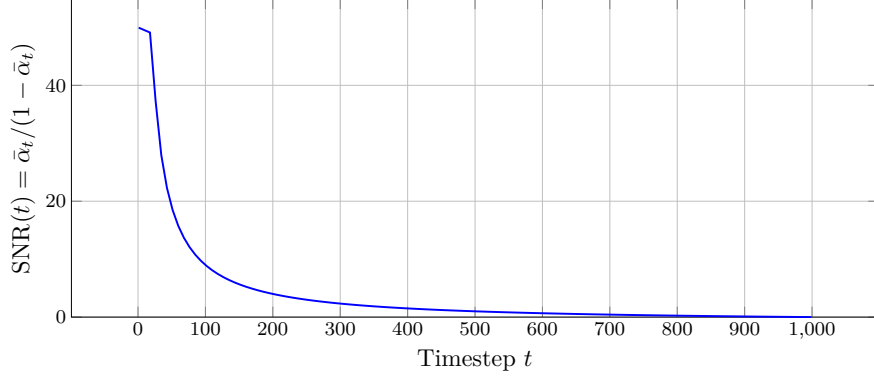
Interpolates conditional and unconditional noise predictions:

$$\tilde{\epsilon}_{\theta} = \epsilon_{\theta}(\mathbf{z}_t, t, \emptyset) + w (\epsilon_{\theta}(\mathbf{z}_t, t, \mathbf{c}) - \epsilon_{\theta}(\mathbf{z}_t, t, \emptyset)). \quad (5)$$

$w > 1$ sharpens samples at cost of diversity.

Table 1: Pixel vs. latent diffusion.

Aspect	Pixel DDPM	Latent (SD)
Resolution	Up to 256^2	512^2 to 1024^2
VRAM	High	Moderate
Text conditioning	Add-on	CLIP + cross-attn

**Figure 2:** Schematic SNR decay when $\bar{\alpha}_t$ decreases approximately linearly with t (see Module 07).

7 Lab

Lab 08 trains a DDPM on MNIST using (1); compare sample quality to GANs in **Module 06** (DCGAN guidelines).

8 Noise Schedules in Practice

Linear β_t schedule from **Module 07** (noise schedule) works for DDPM; latent models often use cosine. Classifier guidance adds $\nabla_{\mathbf{x}_t} \log p(c | \mathbf{x}_t)$ to the score during sampling.

8.1 EMA Weights

Maintain exponential moving average $\theta_{\text{EMA}} \leftarrow 0.999 \theta_{\text{EMA}} + 0.001 \theta$ for stabler samples.

9 Extended Intuition

DDPM implements discrete reverse steps that invert the forward noising process learned during training. At inference you start from $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and iterate $t = T, \dots, 1$ using the network’s noise predictions. Each reverse step adds a small amount of Gaussian noise (except often at $t = 1$) to match the learned transition kernels.

The simplified training objective from Ho et al. is mean squared error on ϵ :

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} \left[\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2 \right]. \quad (6)$$

Sampling uses ϵ_θ to estimate denoised direction; DDIM and other samplers reduce step count for deployment.

Stable Diffusion runs diffusion in a VAE latent space ($\mathbf{z} \in \mathbb{R}^{4 \times 64 \times 64}$ for 512×512 images) so attention cost drops. Classifier-free guidance interpolates conditional and unconditional noise predictions at sample time for sharper text-conditioned images. Fine-tuning Stable Diffusion on a

1650 is usually impractical for full weights; LoRA adapters from **Module 11** are the practical path.

9.1 Reverse Process and Variance

Ho et al. fix reverse variance to $\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$ (posterior variance under Gaussian assumptions). Learned variance (Improved DDPM) adds channels to the network output; helpful at high resolution but more tuning on MNIST-scale labs. DDIM treats reverse steps as non-Markovian with noise scale η ; $\eta = 0$ gives deterministic sampling, $\eta = 1$ matches DDPM stochasticity. Respacing: train with $T = 1000$ but sample with 50 evenly spaced timesteps by jumping $\bar{\alpha}_t$ indices; quality often holds on simple datasets. Probability flow ODE (advanced) treats reverse diffusion as an ODE with $\eta = 0$ DDIM; useful mental model when reducing steps below 50 on MNIST demos. Improved DDPM learns diagonal variance in $\mathbf{x}_0$ prediction; skip unless basic MNIST sampling already looks sharp at 1000 steps.

10 Numerical Walkthrough

MNIST DDPM: 28×28 , $T = 1000$, U-Net base channels 64, $B = 64$, lr = 2×10^{-4} .

Training step: sample batch images $\mathbf{x}_0$, random $t \in \{0, \dots, 999\}$, MSE on predicted vs true noise. Typical initial loss ≈ 1.0 (unit variance noise in 784 dims per pixel channel averaged) dropping toward 0.05 to 0.15 when converging.

Reverse step (conceptual, DDPM):

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \alpha_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}. \quad (7)$$

Full sampling = 1000 network evaluations per image (slow on 1650: minutes per sample without step reduction).

DDIM with 50 steps: subsample timesteps, deterministic update ($\eta = 0$), quality often acceptable on MNIST for labs. Latent SD: VAE encodes $512 \times 512 \rightarrow 64 \times 64 \times 4$; diffusion U-Net operates on 64px grid with cross-attention to text embeddings.

Classifier-free guidance at inference: $\tilde{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \emptyset) + w(\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, c) - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \emptyset))$ with guidance weight $w \approx 7.5$ for SD. Training for guidance: randomly drop text condition to empty prompt 10 to 20% of steps so the model learns unconditional denoising alongside conditional. v-prediction parameterization (advanced) targets velocity $\mathbf{v} = \sqrt{\alpha_t}\boldsymbol{\epsilon} - \sqrt{1 - \alpha_t}\mathbf{x}_0$; some schedulers prefer it at low noise levels.

Sampling step count dominates wall time: 1000 U-Net forwards at 28×28 on a 1650 may take 2 to 5 minutes per image batch; profile one reverse loop before training 200k steps. Classifier guidance (deprecated in text-to-image) adds $\nabla_{\mathbf{x}_t} \log p(c | \mathbf{x}_t)$ to the score; classifier-free guidance replaced it by training without labels intermittently. Latent diffusion trades spatial resolution for channel depth; VAE scale factor 0.18215 appears in SD pipelines and must match when decoding samples. PLMS and DPM-Solver reduce steps below DDIM on some checkpoints; stick to DDIM in the base lab before chasing sampler variants.

11 PyTorch Implementation Notes

- U-Net: downsampling conv blocks with skip connections, timestep injection via `nn.Embedding` + linear projection into feature maps.
- `diffusers` library: `DDPMPipeline`, `UNet2DModel`, `DDPMScheduler` for reference implementations.

- Sampling loop: `model.eval()`, `torch.no_grad()`, iterate scheduler timesteps in reverse order.
- For MNIST lab, 1-channel U-Net; Stable Diffusion checkpoints expect 4-channel latent input and CLIP text encoder outside U-Net.
- EMA weights (exponential moving average of model params) often improve sample quality; update every step with decay 0.9999.

```
for t in scheduler.timesteps:
    eps = model(x, t).sample
    x = scheduler.step(eps, t, x).prev_sample
```

12 Local GPU Constraints

MNIST DDPM U-Net, $B = 64$, trains on 4 GB with fp32; sampling 1000 steps is the bottleneck, not training memory. Use DDIM 50 to 100 steps for demo grids during development.

Full Stable Diffusion inference (fp16) needs 6 to 8 GB+; on 1650 use `enable_attention_slicing`, `enable_vae_slicing`, or CPU offload from `diffusers`. DreamBooth/LoRA fine-tune rank-4 adapters on small image sets at 512 px with $B = 1$, gradient accumulation 8+.

13 Debugging Checklist

- Gray mush samples: insufficient training steps or wrong image scaling ($[0, 1]$ vs $[-1, 1]$).
- NaN during sampling: fp16 overflow; force fp32 in sampler or clamp $\mathbf{x}_t$.
- Periodic vertical stripes: U-Net alignment bug in skip connections (shape mismatch padded incorrectly).
- SD out-of-memory: reduce resolution to 384, use xFormers/memory efficient attention if available.
- Text conditioning ignored: classifier-free guidance scale 1.0; try 7.5 at inference for SD pipelines.
- Sample grid identical across runs: forgot to fix RNG seed for initial $\mathbf{x}_T$; each sample should start from independent noise.

14 Practice Problems

Problem 1. Compare sample quality and wall time for 1000-step DDPM vs 50-step DDIM on the same checkpoint.

Problem 2. Implement EMA weight tracking and generate samples from EMA vs raw weights.

Problem 3. Estimate VRAM for U-Net forward at 28×28 vs 64×64 with base channels 128.

Problem 4. Run pretrained Stable Diffusion with guidance scale 1, 7.5, 15 and note artifact tradeoffs.

Problem 5. Implement respaced sampling with 50 timesteps from a 1000-step trained model. Compare FID or visual sharpness to full 1000-step chain.

Problem 6. Log MSE per timestep bucket ($t \in [0, 333), [333, 666), [666, 999]$). Which range dominates error at epoch 10 vs epoch 50?

U-Net skip connections must match spatial dims exactly; off-by-one upsample padding causes crop artifacts in generated MNIST digits. Save intermediate $\mathbf{x}_t$ grids every 100 reverse steps during

sampling; stuck noise patterns often mean wrong σ_t or scheduler index. Compare EMA vs raw weights on the same noise seed before declaring sampling broken; EMA often fixes speckle without retraining. When running SD demos on 1650, enable sequential CPU offload before lowering resolution below 384px. DDIM η interpolates stochasticity: sweep $\eta \in \{0, 0.5, 1\}$ on MNIST samples after training converges. Profile one full reverse loop before scaling training; sampling not forward pass usually limits DDPM lab turnaround.

Train DDPM in **Lab 08**; compare GAN samples from **Module 06**. Adapter tuning connects to **Module 11**.

Source certification

This module lists where each core definition, equation, figure, and summary table comes from. Pedagogical simplifications (lab batch sizes, epoch counts, illustrative plots) are labeled in extension sections and are not claimed as optimal production settings.

Table 3: Certified topic map for Module 08.

Topic	Primary source	Where in this PDF
DDPM simplified training loss	[1]	Eq. 2
Reverse sampling steps	[1]	Section 3
Latent diffusion (Stable Diffusion)	[3]	Section 5
Improved schedules / samplers	[2]	Section 8

Full bibliographic entries appear below. Figures are schematic unless a caption cites a specific paper figure. If a statement is not listed here, treat it as general ML practice or lab-specific guidance, not a cited theorem.

References

- [1] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *NeurIPS*, 2020.
- [2] Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *ICML*, 2021.
- [3] Robin Rombach, Andreas Blattmann, Dominik Lorenz, et al. High-resolution image synthesis with latent diffusion models. In *CVPR*, 2022.
- [4] Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *ICML*, 2015.
- [5] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, et al. Score-based generative modeling through stochastic differential equations. In *ICLR*, 2021.